

Turbulence Modeling Resource

for the validation and verification of turbulence models

Important Note

As of 02/24/2026, the entire TMR site has been relocated to **GitHub**; the original NASA Langley URL now redirects to a page that points to this new site. This action has been taken to facilitate ongoing maintenance and productivity for the TMR. Most everything should have been carried over relatively seamlessly to the new location.

- New site: <https://tmbwg.github.io/turbmodels/index.html> <- please bookmark this page!
- GitHub repository: <https://github.com/TMBWG/turbmodels>

If you have a link to a specific page on the old TMR website, you can find the equivalent page on the new TMR by replacing "turbmodels.larc.nasa.gov/" with "tmbwg.github.io/turbmodels/" in the hyperlink. For instance, the link:

"https://turbmodels.larc.nasa.gov/Other_LES_Data/nasa_hump_2009.html" should be changed to "https://tmbwg.github.io/turbmodels/Other_LES_Data/nasa_hump_2009.html"

Purpose

The purpose of this site is to provide a central location where Reynolds-averaged Navier-Stokes (RANS) turbulence models are documented. This effort is guided by the [Turbulence Model Benchmarking Working Group](#) (TMBWG), a working group of the [Fluid Dynamics Technical Committee](#) of the [American Institute of Aeronautics and Astronautics](#) (AIAA).

The objective is to provide a resource for CFD developers to:

- obtain accurate and up-to-date information on widely-used RANS turbulence models, and
- verify that models are implemented correctly.

This latter capability is made possible through "verification" cases. This site provides simple test cases and grids, along with sample results (including grid convergence studies) from one or more previously-verified codes for some of the turbulence models. Furthermore, by listing various published variants of models, this site establishes naming conventions in order to help avoid confusion when comparing results from different codes.

The site should also help CFD code users to understand and compare the predictions of a variety of models on the fundamental flow problems in the validation database. Note that it is not the intention of this effort to provide validation of turbulence models for a wide range of complex flows for diverse applications. While this would undoubtedly be valuable, it is beyond the scope of what can be supported. Instead, the goal is to provide a set of test cases that illustrate the performance

of models for flows that capture fundamental phenomena, in order to establish a consistent basis of comparison as a starting point from which a more thorough validation effort for flows of specific interest to users and developers can be conducted.

The site also serves the turbulence modeling community in other ways. It provides a storehouse of turbulent flow validation databases from various sources, including experiment, DNS, and LES. It includes some information about turbulent manufactured solutions. It provides additional cases and grids specifically for use in workshops focused on turbulence model numerical analysis.

Finally, the site should serve as a forum for model developers to help disseminate new models to the CFD community.

If you have any suggestions regarding things that could be added to this website, please email the Page Curator(s) listed at the bottom of the page. Suggestions may also be given to the following people in the TMBWG: [Brian Smith](#) of Lockheed-Martin, or [George Huang](#) of Wright State University.

[NOTES ON HOW TO USE THIS WEBSITE](#)

Turbulence Models

- One-Equation Models:
 - [Spalart-Allmaras](#)
 - [Nut-92](#)
 - [Wray-Agarwal](#)
- Two-Equation Models:
 - [Menter k-omega GEKO](#)
 - [Menter k-omega SST](#)
 - [Menter k-omega BSL](#)
 - [Wilcox k-omega](#)
 - [Chien k-epsilon](#)
 - [K-kL](#)
 - [Explicit Algebraic Stress k-omega](#)
 - [Explicit Algebraic Stress k-kL](#)
- Three-Equation Models:
 - [K-e-Rt](#)
- Three-Equation Models plus Elliptic Relaxation:
 - [K-e-zeta-f](#)
- Seven-Equation Omega-Based Full Reynolds Stress Models:
 - [Wilcox Stress-omega](#)
 - [SSG/LRR](#)
- Seven-Equation Epsilon-Based Full Reynolds Stress Models:
 - [GLVY Stress-epsilon](#)

Turbulence+Transition Models

- One-Equation Models:
 - [SA-BCM](#)
 - [WA-AT](#)
- Two-Equation Models:
 - [K-gamma](#)
 - [WA-gamma](#)
- Three-Equation Models:
 - [SA-AFT](#)
 - [SA-noft2-Gamma-Retheta](#)
 - [SST-2003-Menter-Gamma-2015](#)
- Four-Equation Models:
 - [SST-2003-LM2009](#)
- Nine-Equation Models:
 - [SSG/LRR-trans](#)

(Guidelines for submitting a new turbulence model description: [Guideline-turbmodeldescription.pdf](#))

[Implementing Turbulence Models into the Compressible RANS Equations](#)

[Notes on running the cases with CFD](#)

Turbulence Model Verification Cases and Grids

- VERIF/2DZP: [2D Zero pressure gradient flat plate](#)
- VERIF/2DCJ: [2D Coflowing jet](#)
- VERIF/2DB: [2D Bump-in-channel](#)
- VERIF/2DANW: [2D Airfoil Near-Wake](#)
- VERIF/2DMEA: [2D Multielement Airfoil](#)
- VERIF/3DB: [3D Bump-in-channel](#)

Models Currently Applied to Verification Cases:

(click on number in parentheses in this table to go to the particular validation page)

	MRR	VERIF/2DZP	VERIF/2DCJ	VERIF/2DB	VERIF/2DANW
SA	4	(6)	(3)	(8)	(4)
SA-neg*	4				
SA-QCR2000	3	(2)		(2)	
SA-QCR2013	2	(3)		(2)	
SA-QCR2013-V	1			(2)	
SA-RC	3			(3)	
SA-neg-RC*	3				

SA-RC-QCR2000	3	(3)		(3)	
SA-RC-QCR2013	1	(3)			
SSTm	3			(4)	(2)
SST-Vm	3	(5)	(3)	(3)	
SST-2003m	2			(3)	
BSLm	2	(2)		(2)	
SSG/LRR-RSM-w2012	3	(3)	(3)	(3)	
Wilcox2006-klim-m	2	(2)	(2)	(2)	
k-kL-MEAH2015m	3	(2)	(2)	(3)	

*SA-neg results should be essentially identical to SA, and SA-neg-RC results should be essentially identical to SA-RC

(Number of codes included in verification are indicated by the value in parentheses)

Additional Turbulence Model Verification Cases and Grids

Used in past AIAA-sponsored workshops (this section formerly titled "Cases and Grids for Turbulence Model Numerical Analysis")

Currently includes only SA or SA-QCR2000 verification cases

- [2D Finite Flat Plate](#) (SA)
- [2D NACA 0012 Airfoil](#) (SA)
- [3D Modified Bump](#) (SA-neg)
- [3D Modified Supersonic Square Duct](#) (SA-QCR2000)
- [2D Hemisphere Cylinder](#) (never completed)
- [3D Hemisphere Cylinder \(old\)](#) (SA-neg)
- [3D Hemisphere Cylinder \(new\)](#) (SA-neg)
- [3D ONERA M6 Wing](#) (SA-neg)

Turbulence Model Validation Cases and Grids

- **Basic Cases:**
 - 2DZP: [2D Zero pressure gradient flat plate](#)
 - 2DML: [2D Mixing Layer](#)

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• Extended Cases:

- **Extended Cases:**
 - 2DZPH: [2D Zero pressure gradient high Mach number flat plate](#)
 - 2DBFS: [2D Backward facing step](#)
 - 2DN44: [2D NACA 4412 airfoil trailing edge separation](#)
 - 2DCC: [2D Convex curvature boundary layer](#)
 - 2DWMH: [2D NASA wall-mounted hump separated flow](#)
 - ACC-SBLI: [Axisymmetric Compression Corner Interaction near Mach 7](#)
 - ACSSJ: [Axisymmetric Cold Supersonic Jet](#)
 - AHSSJ: [Axisymmetric Hot Supersonic Jet](#)
 - AJM163TM: [Temperature-Matched Mach 1.63 Axisymmetric Jet](#)
 - AJM163H: [Heated Mach 1.63 Axisymmetric Jet](#)
 - AJM163OD: [Off-Design Mach 1.63 Axisymmetric Jet](#)
 - 3DSSD: [3D Supersonic square duct](#)
 - AISBLI: [Axisymmetric Impinging-Shock Boundary-Layer Interaction](#)

Models Currently Applied to Validation Cases:

(click on number in parentheses in this table to go to the particular validation page)

[illegible]

SSG/LRR-RSM-w2012	3	(2)	(2)	(2)	(3)				(2)	(2)		
Wilcox2006-klim-m	2	(2)	(2)	(2)	(2)				(2)	(2)	(2)	(2)
k-kL-MEAH2015m	3	(2)	(2)	(3)	(3)					(3)		(3)
EASMko 2003-S	1	(1)	(1)	(1)	(1)				(1)		(1)	(1)
k-e-Rt	1	(1)		(1)	(1)				(1)		(1)	(1)
k-e-Rt-RC	1											
GLVY- RSM-2012	1	(1)										

(* indicates "Basic Case")

(Number of codes included in validation are indicated by the value in parentheses)

Alternative Validation Case Listing by Flow Physics:

		Free shear flows			Wall flows		P-grad-ients	Curv-ature	Compressibility		
		Jet	Mixing layer	Wakes	Law of wall	Law of wake			Mixing	Van Driest I	Van Driest II
Boundary layers	2DZP*				Y	Y					
	2DZPH									Y	Y
	ASBL*				Y		weak				
Mixing layer/ wakes	2DML*		Y								
	2DANW*			Y							
Jets	ASJ*	Y									
	ANSJ*	Y							Y		
	AHSJ*	Y									
	ACSSJ	Y							Y		
	AHSSJ	Y							Y		
	AJM163TM	Y							Y		
	AJM163H	Y							Y		
	AJM163OD	Y							Y		
Airfoils	2DN00*						Y				
	2DN44						Y				
Bump flows	ATB*						Y				

	2DWMH						Y				
Shock/boundary layer interaction flows	ACC-SBLI						Y				
	AISBLI						Y				
Internal flows	2DCC						Y	Y			
	2DBFS						strong				
	3DSSD						Y				

(* indicates "Basic Case")

"Classical" Turbulence Model Behavior at Very High Reynolds Number

- 2DFDC: [2D Fully-Developed Channel Flow at High Reynolds Number](#)

Transition Model Verification Cases and Grids



UNDER CONSTRUCTION

- VERIF/2DTFP: [2D T3A Transitional Flat Plate](#)

Turbulent Flow Validation Databases

The data in the following links are publicly available and are provided here as a convenience. They are provided as-is and accuracy is not guaranteed; questions should be directed to the sources of the data provided.

- [Data from "Collaborative Testing of Turbulence Models"](#)
- [Data from Experiments](#)
- [Data from Direct Numerical Simulations \(DNS\)](#)
- [Data from Large Eddy Simulations \(LES\)](#)

Turbulent Manufactured Solutions

- [Information from Lisbon "Workshop on CFD Uncertainty Analysis" series](#)

Related Workshops (in reverse chronological order)

- [NASA High Fidelity CFD Workshop 2025](#) (concluded)
- [Symposium on Turbulence Modeling: Roadblocks, and the Potential for Machine Learning 2022](#) (concluded)
- [High Fidelity CFD Workshop 2022](#) (concluded)
- [NASA's 40% Challenge and CFD Prediction Error Assessment Workshop 2018](#) (concluded)
- [NASA Transition Modeling Workshop 2017](#) (concluded)

[Model Readiness Rating System \(MRR\)](#)

Related External Links to Validation Databases and Other Resources

- [ERCOFTAC Database](#)
- [ERCOFTAC Knowledge Base Wiki](#)
- [JAXA DNS Database](#)
- [Johns Hopkins Turbulence Databases \(JHTDB\)](#)
- [Kawai Lab - DNS/LES Database](#)
- [KTH Boundary Layer DNS/LES Data](#)
- [NASA Common Research Model](#)
- [NPARC Alliance CFD Verification and Validation](#)
- [University of Maryland Wall-Modeled Large Eddy Simulation Resource](#)

Related External Links to CFD Codes

- [CFL3D structured grid code](#)
 - [FUN3D unstructured grid code](#)
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Recent Significant Site Updates

07/30/2026 - Added Axisymmetric Impinging SBLI case; renamed Mach 7 compression corner SBLI case

03/03/2026 - Added the Generalized K-Omega Model by Menter and Matyushenko 02/24/2026 - Entire TMR site relocated to GitHub; original NASA Langley site now redirects to a page that points to this new site

----- New site: <https://tmbwg.github.io/turbmodels/index.html>

----- GitHub repository: <https://github.com/TMBWG/turbmodels>

06/15/2024 - Renamed "Cases and Grids for Turbulence Model Numerical Analysis" and moved closer to Verification Cases

07/26/2021 - Added external link to JAXA DNS Database site

03/24/2021 - clarifications on use of "m" designation when $P=\mu_t S^2$ and k term ignored in momentum and energy equations in 2-equation models (throughout site)

11/12/2020 - Added description of SA-AFT 3-eqn turbulence+transition models

08/28/2020 - introduced SST-Vm variant of SST, and changed SST-V naming to SST-Vm on many of the results pages

07/20/2020 - Added [SA-BCM](#) transition model description

06/04/2019 - Added [NASA Juncture Flow \(JF\)](#) data

Page Curators: [Ethan Vogel](#), [Clark Pederson](#)

Last Updated: 07/07/2026